

ORIGINAL RESEARCH

HEART FAILURE

Effects of Atrial Fibrillation Ablation for Heart Failure With Preserved Ejection Fraction

Insights From CABANA



Pieter Martens, MD, MSc, PhD,^{a,b} Silvio Nunes Augusto Jr,^c Jonas Erzeel, MD,^{a,b} Laurent Pison, MD, PhD,^a Wilfried Mullens, MD, PhD,^{a,b} W.H. Wilson Tang, MD^{c,d}

ABSTRACT

BACKGROUND Atrial fibrillation (AF) ablation is Class I recommendation in selected heart failure (HF) patients with reduced ejection fraction; less is known in heart failure with preserved ejection fraction (HFpEF).

OBJECTIVES The aim of this study was to investigate the effects of AF ablation in patients with HFpEF.

METHODS The CABANA (Catheter Ablation vs Antiarrhythmic Drug Therapy for Atrial Fibrillation) trial randomized patients with cardiovascular risk factors for stroke to AF ablation vs drug therapy. The presence of a high likelihood of HFpEF at enrollment was determined by a modified H₂FPEF score of ≥ 6 . Treatment effects of baseline HFpEF likelihood on the AF ablation for death and cardiovascular admission, AF recurrence, and functional status were assessed.

RESULTS A total of 1,763 patients were included in the analysis. A high modified H₂FPEF score (55% of the entire cohort) resulted in a significant treatment effect modulation (P for interaction = 0.027), with a lower risk for cardiovascular hospitalization or death in patients with a high likelihood of HFpEF (HR: 0.82 [95% CI: 0.69-0.98]; P = 0.025), but not in patients without (HR: 1.00 [95% CI: 0.82-1.22]; P = 0.987). Although patients with a high likelihood of HFpEF were at a higher risk for AF recurrence, the greatest treatment effect of AF ablation on AF recurrence was observed in patients with a high likelihood of HFpEF (P for interaction = 0.035). In a sensitivity analysis in a subset of patients with echocardiographic evidence of HFpEF (n = 225), a similar treatment interaction was found.

CONCLUSIONS In patients undergoing AF ablation, the presence of underlying HFpEF (either by HFpEF probability or defined by echocardiography) was associated with a larger benefit with AF ablation on clinical outcome, AF recurrence, and functional status. (Catheter Ablation vs Antiarrhythmic Drug Therapy for Atrial Fibrillation [CABANA]; [NCT00911508](https://clinicaltrials.gov/ct2/show/study/NCT00911508)) (JACC Heart Fail. 2025;13:785-794) © 2025 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

From the ^aDepartment of Cardiology, Ziekenhuis Oost-Limburg A.V., Genk, Belgium; ^bHasselt University, Diepenbeek/Hasselt, Belgium; ^cCardiovascular and Metabolic Sciences, Lerner Research Institute, Cleveland Clinic, Cleveland, Ohio, USA; and the ^dKaufman Center for Heart Failure Treatment and Recovery, Department of Cardiovascular Medicine, Heart Vascular and Thoracic Institute, Cleveland Clinic, Cleveland, Ohio, USA.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received October 2, 2024; revised manuscript received January 20, 2025, accepted January 23, 2025.

**ABBREVIATIONS
AND ACRONYMS****AF** = atrial fibrillation**HF** = heart failure**HFpEF** = heart failure with
preserved ejection fraction**HFrfEF** = heart failure with
reduced ejection fraction**LVEF** = left ventricular ejection
fraction

Catheter ablation for atrial fibrillation (AF) has emerged as a Class I recommendation for selected patients with heart failure with reduced ejection fraction (HFrfEF), HYP¹ based on the results of several randomized controlled trials showing a lower risk for heart failure (HF) admissions and cardiovascular mortality after catheter ablation.²⁻¹¹ Less is known about the effect of AF ablation in patients with heart failure with preserved ejection fraction (HFpEF). A recent randomized controlled trial in patients with AF and HFpEF illustrated that AF ablation resulted in a lower capillary pulmonary wedge pressure at follow-up.¹² Nevertheless, data on the effect of AF ablation in patients with AF and HFpEF on hard clinical outcome are lacking, in contrast to HFrfEF. This despite the knowledge that AF and HFpEF often coexist and both diseases share mutual risk factors in their development (indirect effect of comorbidities or epicardial fat directly) and share morphological and functional abnormalities that are associated with disease progression, such as left atrial dilation and left atrial dysfunction.¹³ More recently, it is becoming increasingly recognized that HFpEF is underdiagnosed in AF, because the signs and symptoms of AF and HFpEF often overlap, and the investigator-reported prevalence of HFpEF is often low in AF trials. This is in contrast to the high HFpEF prevalence of $\leq 73\%$ when invasive hemodynamic criteria for HFpEF are applied to AF patients undergoing AF ablation.¹⁴ Signs and symptoms of AF and HFpEF often overlap. In contrast with invasive hemodynamics, the H₂FPEF score is an approach that allows for the estimation of the clinical probability of HFpEF without the need of more cumbersome invasive hemodynamics and does not require biomarker data (eg, N-terminal pro-B-type natriuretic peptide), such as is the case for the HFA PEF score.¹⁵ The H₂FPEF score is widely used in clinical practice and has an excellent diagnostic capability (a score of ≥ 6 has a $>90\%$ probability for HFpEF) and prognostic usefulness.¹⁶ More recent data demonstrate that applying the H₂FPEF score to newly diagnosed AF patients documents a HFpEF prevalence of 60%.¹⁷ The goal of our analysis was 2-fold: 1) to determine the prevalence of a high likelihood of HFpEF in patients enrolled in the CABANA (Catheter Ablation vs Antiarrhythmic Drug Therapy for Atrial Fibrillation; [NCT00911508](#)) trial using the H₂FPEF score; and 2) to determine the potential effect of AF ablation in patients with vs without a high likelihood of HFpEF at baseline enrolled in the CABANA trial, on hard endpoints, risk for AF recurrence, and functional status.

METHODS

TRIAL DESIGN AND PATIENT POPULATION. The design and the results of the CABANA trial have been published previously.^{18,19} In brief, patients ≥ 65 years or < 65 years with 1 risk factor for stroke (hypertension, HF, history of stroke, diabetes or other heart problems) who had ≥ 2 episodes of paroxysmal AF or 1 episode of persistent AF in the prior 6 months were eligible for randomization. Patients were excluded if they had prior AF ablation or failed ≥ 2 antiarrhythmic drugs. Patients were randomized in a 1:1 fashion to either AF ablation or drug therapy (consisting of either rhythm and/or rate control) with stratification for site. At each site, the institutional review board or ethics committee approved the study, and all patients provided written informed consent before study enrollment. The data from the CABANA trial was accessed through the National Heart Lung and Blood Institute's BioLINCC (Biologic Specimen and Data Repository Information Coordinating Center). This report complies with the declaration of Helsinki and is written according to the STROBE guidelines for randomized controlled trials. This study is approved by the Cleveland Clinic Institutional Review Board.

INTERVENTION. The CABANA trial compared 2 treatment strategies, namely, drug therapy or AF ablation. The goal of the drug therapy arm consisted of controlling AF rhythm or rate. The goal of the AF ablation arm was to eliminate AF. In the drug arm, patients underwent first rate control to achieve resting heart rates < 80 beats/min and exercise heart rates on 6-minute walk test or Holter monitoring of < 110 beats/min. In case of failure to achieve this rate control target or in the setting of persistent symptoms despite achieving rate control targets, patients in the drug arm were eligible to receive rhythm control with an antiarrhythmic drug (choice of agent depending on patient profile) for an appropriate period and cardioverted to sinus rhythm if necessary. Ablation of the atrioventricular conduction system was allowed in case of inadequate rate control refractory to ≥ 3 rate control agents or 2 antiarrhythmic drugs. All patients in the AF ablation arm underwent pulmonary vein isolation with the choice of the used guiding system and the use of additional ablative strategies being left at the discretion of the treating physician.

HF STATUS AT BASELINE. We used a modified H₂FPEF score to establish a clinical probability of HFpEF at baseline in all patients with a baseline left ventricular ejection fraction (LVEF) $> 50\%$. This is a numeric score ranging from 0 to 9 with a score of ≥ 6

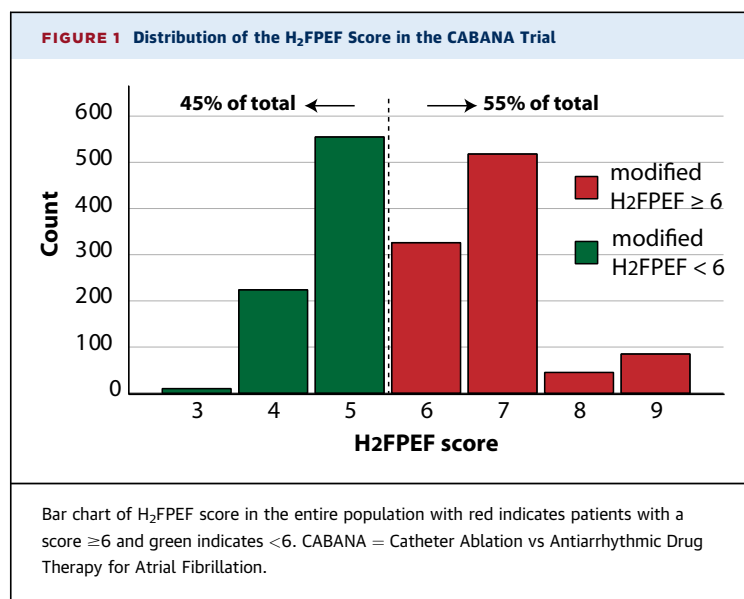
being diagnostic for HFpEF. This score is validated against the gold standard of invasive hemodynamics.^{15,16} Components of the score include the use of ≥ 2 antihypertensive agents, Doppler pulmonary artery systolic pressure of >35 mm Hg, age >60 years, and Doppler E/e' ratio of >9 generating a total weight of 1, body mass index of >30 kg/m² generate a weight of 2 and presence of AF generates a weight of 3, giving a total score of 9. For echocardiographic parameters, we used the baseline diastolic function assessment in CABANA reported as normal/grade 1 vs ≥ 2 , this because E/e' ratio and Doppler pulmonary artery systolic pressure of >35 mm Hg were not available, and therefore we refer to it as a modified H₂FPEF score. We allocated 0 points to normal or diastolic grade 1 and allocated 2 points to a grade ≥ 2 . All echocardiographic measurements were performed and reported by the sites. This trial did not have a central core lab reading of baseline echocardiographic data. In line with the published reports a score of ≥ 6 was used to define HFpEF at baseline ($>90\%$ probability for HFpEF confirmed by invasive hemodynamics). Patients with a score between 4 and 5 can still have HFpEF, but often need additional testing to confirm a diagnosis (eg, exercise hemodynamics). For simplicity, those patients were also defined as no clear HFpEF at baseline in the remainder of this paper.

SENSITIVITY ANALYSIS. Given the strong reliance of the H₂FPEF score on AF (contribution to 3 points), we performed a sensitivity analysis looking at patients with echocardiographic evidence of HFpEF at baseline. This defined as symptoms of HF in addition to structural abnormalities that fulfill morphological criteria for HFpEF including left atrial dilation (left atrial diameter of >40 mm or left atrial volume index of >34 mL/m²) and left ventricular hypertrophy by increased ventricular thickness (either septum or posterior wall thickness above the sex specific normal value cutoff).²⁰ Given the overlap of symptoms of HF and AF, symptoms was defined as either a NYHA functional class or CCS (Canadian Cardiovascular Society) class of ≥ 1 , patients with no symptoms were not included in the sensitivity analysis. Patients with both symptoms of HF and the presence of both left atrial dilation and left ventricular hypertrophy were defined as having echocardiographic evidence of HFpEF. The interaction between baseline echocardiographic evidence of HFpEF and the treatment effect of AF ablation on the primary endpoint the risk for all-cause mortality and cardiovascular hospitalization was assessed as a sensitivity analysis.

ENDPOINTS. To assess the effect of AF ablation in patients with HFpEF, cardiocentric endpoints relevant to HFpEF in line with the original report of the CABANA trial were selected. Trial endpoints include: 1) the risk for all-cause mortality and cardiovascular hospitalization; 2) the risk for all-cause mortality; 3) the risk for AF recurrence at follow-up; and 4) functional status at follow-up, including the NYHA functional class and CCS Severity of Atrial Fibrillation score. Additionally, weight change after randomization was assessed as a surrogate of improving HF status. For AF recurrence we used simplified approach of looking at all postrandomization electrocardiograms available according to the randomized arm and baseline H₂FPEF score category and AF recurrence was defined as AF documented on any full electrocardiogram. This approach does not take a blanking period or the start/stop of antiarrhythmic drugs into account.

STATISTICAL ANALYSIS. Baseline characteristics are summarized as mean $\pm$ SD, median (25th-75th percentile), or numbers and percentages and were evaluated using the independent samples Student's *t*-test, Mann-Whitney U test, or chi-square test as appropriate. The combined endpoint of all-cause mortality and cardiovascular hospitalization was assessed in time-to-event analysis using a Cox proportional hazard model including the treatment arm and interaction with baseline HF₂PEF score of ≥ 6 or <6 to calculate HRs and 95% CI and reporting a *P* value for interaction. Nonproportionality of hazards among treatment arms was assessed by adding an interaction of treatment groups with the logarithm of time to the Cox proportional hazard model.

The binary endpoint of AF recurrence at follow-up was evaluated using a generalized linear mixed model (models equipped for response variables of different nature, including binary endpoints or continuous endpoints) in a model allowing for repeated measurements. The generalized linear mixed model included a fixed treatment effect and random intercept to calculate ORs and 95% CIs (logit link function). For interaction analysis with baseline HF₂PEF score of ≥ 6 or <6 was entered into the model as a fixed effect interaction-term with treatment allocation. Change in NYHA functional class, CCS class, and weight after randomization were assessed using a linear mixed effect model for repeated measurements with a fixed treatment effect and the interaction with baseline HF₂PEF score of ≥ 6 or <6 and a random intercept. No multiplicity adjustments were done for all secondary analysis, so all reported values are exploratory. Hypotheses testing is 2-sided and a significance level of $\alpha = 0.05$ was used;



however, treatment interaction values of <0.1 were assumed to indicate treatment effect modulation. All statistical analyses were performed according to the intention to treat principle and were performed using SPSS version 25 or RStudio 2022.02.3+492.

RESULTS

PATIENT POPULATION AND RISK ASSOCIATED WITH A HIGH HFpEF PROBABILITY. A total of 2,204 patients were enrolled in the CABANA trial. The modified H₂FPEF score could be calculated in a total of 1,763 patients. In 441 patients data was missing not allowing to calculate the H₂FPEF score (exclusively related to the missing of pivotal echocardiographic data.) **Figure 1** illustrates the distribution of the modified H₂FPEF score in the entire population. A total of 55% (of the 1,763 patients) had a score of ≥6. **Table 1** illustrates the baseline features of all patients enrolled in the CABANA trial and further after categorization according to a baseline score of <6 (lower probability of HFpEF) or ≥6 (high probability of HFpEF). Patients with a high probability of HFpEF were older; more often White; had a higher body mass index and a higher blood pressure; more often had diabetes, coronary artery disease, and obstructive sleep apnea; and had worse functional status both by the CSS-AF severity class and NYHA functional class. Median LVEF was 59% (95% CI: 53%-64%). As illustrated in **Table 1**, only 4.9% of patients with a high probability of HFpEF received treatment with a mineralocorticoid receptor antagonist (MRA) at baseline and this percentage was identical to patients with a modified H₂FPEF score of <6. Patients with a

high probability of HFpEF had a trend toward a higher risk to be admitted for cardiovascular causes or had a higher risk for all-cause death (HR: 1.13 [95% CI: 0.99-1.30]; *P* = 0.068) and had a higher risk for AF recurrence at follow-up (OR: 1.1 [95% CI: 1.00-1.20]; *P* = 0.001). Assessment of outcome was complete in all 1,763 patients.

EFFECT OF AF ABLATION ON ADVERSE CLINICAL OUTCOMES ACCORDING TO HFpEF PROBABILITY.

Table 2 illustrates the effect of AF ablation on hard clinical outcome in patients with high or lower probability of HFpEF. In the overall population (*n* = 1,763 patients) a total of 904 primary events of cardiovascular hospitalization or all-cause death occurred. In the overall population, AF ablation was not associated with a lower risk for cardiovascular hospitalization or all-cause death. However, baseline HFpEF probability resulted in a significant treatment effect modulation (*P* for interaction = 0.027), with AF ablation resulting in a significant lower risk for cardiovascular hospitalization or all-cause death in patients with a high probability of HFpEF (HR: 0.82 [95% CI: 0.69-0.98]; *P* = 0.025), whereas this effect was not present in patients with a lower probability of HFpEF (HR: 1.00 [95% CI: 0.82-1.22]; *P* = 0.987). No effect of AF ablation was seen on the individual component of all-cause mortality in the overall population or in the subgroups according to baseline HFpEF probability, indicating that the effect on the combined endpoints was driven by a decrease in cardiovascular admissions.

EFFECT OF AF ABLATION ON AF RECURRENCE ACCORDING HFpEF PROBABILITY.

Figure 2 illustrates the effect of AF ablation on AF recurrence. In the overall population, AF ablation resulted in a significant 0.64 lower odds for the development of AF during the entire follow-up. In both patients with a low HFpEF probability (OR: 0.69 [95% CI: 0.63-0.76]; *P* < 0.001) and with a high HFpEF probability at baseline (OR: 0.63 [95% CI: 0.58-0.69]; *P* < 0.001) AF ablation resulted in a lower risk for AF recurrence. However, the treatment effect was larger in patients with high HFpEF probability (*P* for interaction = 0.035), despite the observation that these in general had a higher risk for AF recurrence (OR: 1.1 [95% CI: 1.00-1.20]; *P* = 0.001), indicate that most AF recurrences in patients with a high HFpEF probability occurred in patients randomized toward the drug therapy arm.

EFFECT OF AF ABLATION ON HF STATUS ACCORDING TO HFpEF PROBABILITY. AF ablation resulted in a significant lower NYHA functional class in both patients with and without a high HFpEF probability

TABLE 1 Baseline Characteristics of the Patient Population

	Overall (N = 2,204)	H ₂ FPEF Score <6 (n = 789)	H ₂ FPEF Score ≥6 (n = 974)	P Value
H ₂ FPEF score	6 (5-7)	5 (4-5)	7 (6-7)	<0.001
AF type				0.794
Paroxysmal	745 (42)	332 (42)	413 (42)	
Persistent	842 (48)	374 (47)	468 (48)	
Long persistent AF	176 (10)	83 (11)	93 (10)	
Age, y	67 (62-72)	68 (62-73)	66 (61-71)	<0.001
Female	819 (37.2)	287 (36.4)	389 (39.9)	0.139
Race				<0.001
White	2,032 (92.4)	712 (90.2)	928 (95.3)	
Black	78 (3.5)	22 (2.8)	32 (3.3)	
Other	90 (4.1)	55 (7.0)	14 (1.4)	
BMI, kg/m ²	30.10 (26.51-34.58)	26.60 (24.69-28.65)	33.46 (31.13-37.17)	<0.001
Systolic BP, mm Hg	129 (120-140)	127 (118-139)	130 (120-140)	0.071
Diastolic BP, mm Hg	80 (70-84)	80 (70-84)	80 (70-83)	0.384
LVEF, %	59 (54-64)	59 (53-64)	59 (54-64)	0.524
Hypertension	1,776 (80.6)	576 (73.0)	848 (87.1)	<0.001
Diabetes	561 (25.5)	120 (15.2)	313 (32.1)	<0.001
Sleep apnea	508 (23.1)	114 (14.4)	283 (29.1)	<0.001
Coronary artery disease	424 (19.2)	112 (14.2)	225 (23.1)	<0.001
Family history of AF	252 (11.5)	92 (11.7)	118 (12.1)	0.829
Prior stroke	126 (5.7)	54 (6.8)	54 (5.5)	0.302
Thromboembolic events	90 (4.1)	33 (4.2)	39 (4.0)	0.946
Valve disease	265 (12.0)	100 (12.7)	119 (12.2)	0.829
MRA use	104 (4.7)	39 (4.9)	48 (4.9)	0.989
CCS AF severity class				0.028
0	223 (10.2)	91 (11.6)	80 (8.2)	
1	339 (15.5)	126 (16.0)	144 (14.8)	
2	703 (32.1)	262 (33.3)	314 (32.4)	
3	783 (35.7)	269 (34.2)	359 (37.0)	
4	143 (6.5)	39 (5.0)	73 (7.5)	
NYHA functional class				0.023
≤1	1,408 (64.0)	526 (66.7)	582 (59.7)	
2	776 (35.2)	256 (32.4)	385 (39.5)	
3	18 (0.8)	7 (0.9)	7 (0.7)	

Values are n (%) or median (25th-75th percentile) in case of non-normal distribution of the continuous variable, unless otherwise indicated. P values indicate the difference between patients with a H₂FPEF score of ≥6 or <6.

AF = atrial fibrillation; BMI = body mass index; BP = blood pressure; CCS = Canadian Cardiovascular Society; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist.

($P = 0.007$); however, the value of interaction of $P = 0.068$ suggests a slightly larger effect on NYHA functional class reduction in patients with a high HFpEF probability (Figure 3A). Similarly, AF ablation resulted in a lower CCS AF severity class at follow-up, albeit without clear treatment interaction by HFpEF probability (P for interaction = 0.597) (Figure 3B). Finally, AF ablation resulted in significantly lower weight during follow-up, also without clear treatment interaction by HFpEF-probability (P for interaction = 0.129) (Figure 3C).

EFFECT OF AF ABLATION ON HF STATUS ACCORDING TO ECHOCARDIOGRAPHIC-DEFINED HFpEF. Of the all patients enrolled in the CABANA trial (N = 2,204), 225

patients were symptomatic and had echocardiographic data allowing to determine echocardiographic presence of HFpEF. A flowchart of the generation of this subgroup from the overall cohort is reflected in Supplemental Figure 1. A total of 144 patients (64%) had echocardiographic evidence of HFpEF. In this subgroup we observed a clear treatment interaction between AF ablation and the presence of underlying HFpEF (P for interaction = 0.006) (Table 3) supporting the overall findings.

DISCUSSION

We report novel insights into the impact of HFpEF probability in patients with AF undergoing AF

TABLE 2 Treatment Effect of AF Ablation According to H₂FPEF Status on Hard Outcomes

	Event Rate AF Ablation (n = 893)	Event Rate Drug Therapy (n = 870)	HR	95% CI	P Value	P Value for Interaction
Risk for all-cause death and CV admission						
Overall	426 (48)	478 (55)	0.89	0.76-1.00	0.054	0.027
H ₂ FPEF score <6	192 (47)	206 (53)	1.00	0.82-1.22	0.987	
H ₂ FPEF score ≥6	234 (48)	272 (56)	0.82	0.69-0.98	0.025	
Risk for all-cause death						
Overall	44 (4.9)	53 (6.1)	0.80	0.54-1.19	0.271	0.678
H ₂ FPEF score <6	19 (4.7)	23 (6.0)	0.79	0.43-1.44	0.436	
H ₂ FPEF score ≥6	25 (5.1)	30 (6.1)	0.81	0.48-1.37	0.431	

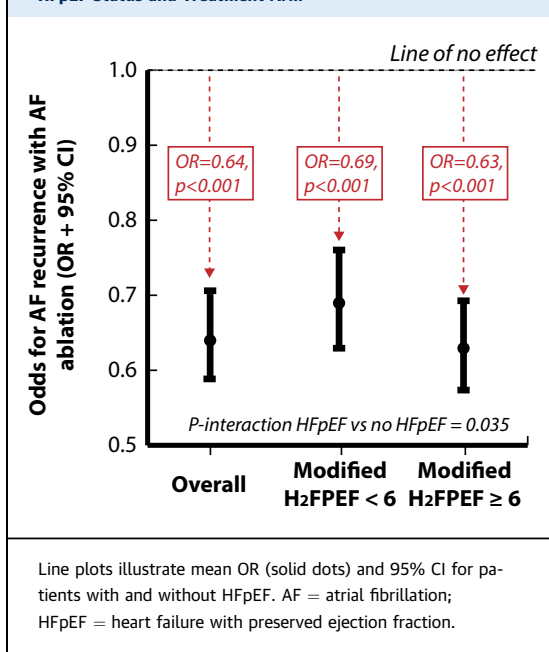
Values are n (%), unless otherwise indicated.
CV = cardiovascular; other abbreviations as in Table 1.

ablation (Central Illustration), with core messages being: 1) the majority of patients undergoing AF ablation have a high probability of HFpEF; and 2) a greater probability of HFpEF is associated with a greater benefit from AF ablation on clinical outcome, AF recurrence, and functional status. The prevalence of investigator-reported HF at baseline in modern AF trials is low (eg, CABANA, ~15%; EAST-AFNET4, ~28%),^{19,21} especially taking into account that these trials were often enriched with cardiovascular risk factors increasing the probability of HF. This result is in contrast to the 55% prevalence of high probability of HFpEF we found in the CABANA trial. As expected, these patients manifested numerous clinical features associated with HFpEF, including White race, sleep

apnea, coronary artery disease, and diabetes in addition to the components that build the H₂FPEF score. These patients with a high HFpEF probability have a higher risk of developing the combined endpoint of all-cause death and cardiovascular readmission and are at increased risk for AF.²² Interestingly, only a minority of patients with HFpEF were treated with MRAs, despite the current recommendation by guidelines in HFpEF, underscoring the potential underdiagnosed burden of HFpEF.²³ Data in HF suggest that MRAs could reduce occurrence of AF.^{24,25} In line with this data, it might therefore be especially valuable to detect HFpEF, because it could lead to better MRA uptake.²⁵ The HFpEF probability scores can easily be applied in clinical practice as it uses readily available clinical variables, however some concerns does exist about the diagnostic accuracy in a sole AF cohort.²⁶ This could help to appreciate the burden of HFpEF and allows for better targeted management.

Next to establishing the potential burden of patients with a high probability for HFpEF in the CABANA trial, our analysis sought to determine the potential treatment benefit of AF ablation in patients with a high probability for HFpEF. In contrast with HFrEF, where a wealth of data from numerous randomized controlled trials illustrated a decrease in HF events and even all-cause mortality,²⁻¹¹ less is known about the effect of AF ablation in HFpEF. Although a recent meta-analysis did potentially suggest less treatment effect with AF ablation in HFpEF, this result only underscores the need for more data.²⁷ Although several observational studies suggested an association between a lower risk of mortality in patients undergoing AF ablation, such studies do not allow any assumptions to be made on causality due to the risk of measured and unmeasured confounding.^{2,4,6,10,11,28} Only well-performed randomized controlled trials allow assumptions to be

FIGURE 2 Risk for Atrial Fibrillation Recurrence According to HFpEF Status and Treatment Arm



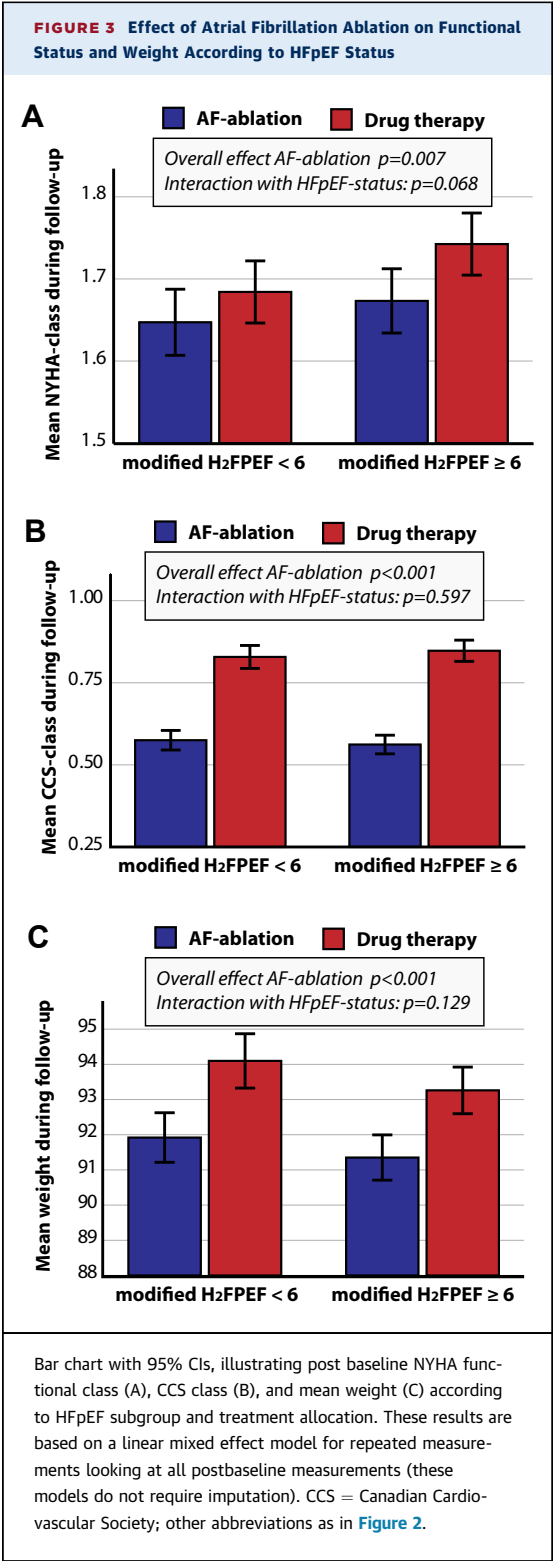


TABLE 3 Sensitivity Analysis for Treatment Effects of AF Ablation According to Echo-Defined HFpEF on Adverse Clinical Outcome

	Event Rate AF Ablation (n = 112)	Event Rate Drug Therapy (n = 113)	HR	95% CI	P Value	P Value for Interaction
Risk for all-cause death and CV admission						
No echo HFpEF (n = 81)	18 (44)	21 (52)	0.93	0.54-1.77	0.829	0.006
Echo HFpEF (n = 144)	39 (54)	44 (61)	0.68	0.44-1.05	0.080	

Values are n (%), unless otherwise noted.
HFpEF = heart failure with preserved ejection fraction; other abbreviations as in Tables 1 and 2.

documented that, in comparison with medical management (n = 15), AF ablation (n = 16) resulted in a lower peak exercise pulmonary capillary wedge pressure and greater improvement in peakVO₂, N-terminal pro-B-type natriuretic peptide, and the Minnesota Living with Heart Failure score.¹² A previous analysis from the CABANA trial restricted to patients with HF suggested a significant benefit of AF ablation in this group of patients.²⁹ However, subgroup analysis in randomized trials should refrain from solely focusing on the subgroup, but should use techniques at baseline to identify randomized patients who do or do not belong to 1 subgroup and subsequently use interaction techniques to determine if the treatment effect is similar or altered in a certain subgroup. In that respect, our analysis adds major information, because it allows us to determine the treatment effect of AF ablation in patients with vs without a high probability for HFpEF. First, our analysis illustrates that AF ablation resulted in a lower combined endpoint of all-cause death and cardiovascular admission in patients with a high probability for HFpEF, whereas this effect was not apparent in patients without a high probability for HFpEF at baseline. This effect seemed to be driven by a significant decrease in cardiovascular admissions in patients with HFpEF. This finding suggests that AF ablation alters the clinical course in patients with HFpEF. This observation is in line with the previous data showing that AF ablation can lower pulmonary capillary wedge pressure in HFpEF patients.¹² Indeed most HF admissions in HFpEF are congestion related, and data from invasive pulmonary pressures monitors illustrate that decreasing the pulmonary capillary wedge pressure decreases the risk for subsequent admissions.³⁰ We also performed a sensitivity analysis in a group of patients with or without echocardiographic evidence of HFpEF and similar results were found. This finding is reassuring, because the

made about the treatment effect of any intervention in a given disease condition. A smaller randomized controlled trial of 31 patients with HFpEF

CENTRAL ILLUSTRATION Effects of Atrial Fibrillation Ablation on HFpEF**Questions**

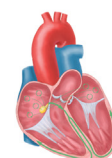
1. What is the proportion of patients with a high probability of HFpEF in AF ablation?
2. What is the treatment effect of AF ablation according to HFpEF probability?

Methods

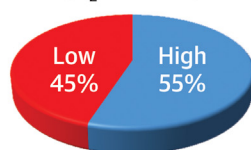
- A. HFpEF probability determined by a modified H₂FPEF score in N = 1,763 patients in CABANA
- B. Interaction analysis: Treatment effect AF ablation × HFpEF probability
- C. Outcomes

Death/CV
admission

Symptoms

AF
recurrence**Results**

Proportion of high
probability of HFpEF
(H₂FPEF ≥ 6)



N = 1,763 with baseline HFpEF probability available of
2,204 patients in the overall trial
randomized to

AF Ablation



Antiarrhythmic Drug

If high HFpEF probability, with AF ablation



Death/CV admission



Symptom improvement



Excellent AF control

Martens P, et al. JACC Heart Fail. 2025;13(5):785-794.

Overview of population and major findings according to HFpEF probability. AF = atrial fibrillation; CV = cardiovascular; HFpEF = heart failure with preserved ejection fraction; R = randomization.

H₂FPEF score does rely heavily on the presence of AF in its calculation.

Second, our analysis illustrates that AF ablation decreases the risk for recurrence of AF at follow-up in comparison with medical management, both in patients with or without a high probability for HFpEF. Although HFpEF is generally accepted as a risk factor for AF recurrence and our analysis also shows that, in general, patients with a high probability for HFpEF are at an increased risk for AF recurrence, this result does not imply that these patients have less of a

treatment benefit from AF ablation. Indeed, patients with a high probability for HFpEF actually had a slightly greater relative treatment benefit from AF ablation (as reflected by the numerical but statistically significant lower odds for AF recurrence at follow-up) in comparison with patients without a high probability for HFpEF.

Finally, our analysis looked at the effect of AF ablation on functional and HF-related metrics. AF ablation resulted in an equal improvement in NYHA functional class and CSS-AF severity class; however,

patients with a high probability for HFpEF did exhibit more symptoms at baseline. This finding perhaps suggests the value of identifying HFpEF in patients undergoing AF ablation. Indeed, AF ablation is often pursued to relieve or improve AF-related symptoms. Patients with a high probability for HFpEF might have different etiologies of an atrial myopathy in addition to just AF itself. Identifying patients with HFpEF might allow for the initiation of disease-modifying therapies that are known to improve HFpEF status. The observation that AF ablation lowers body weight during follow-up gives some biologic credence to the observed reduction in cardiovascular admissions.

STUDY LIMITATIONS. First, due to some missing data at baseline, the H₂FPEF score could not be calculated for all patients. However, the key principle of randomization assures that such missingness occurs balanced in both treatment arms. This entails that the relative treatments effects (OR and HRs in this paper) remain the same. What does change are the CIs, that become broader which increases the risk for type 2 error. Second, an H₂FPEF score of 4 or 5 does not necessarily exclude HFpEF. Third, over the last years novel therapies have emerged for HFpEF, including sodium-glucose cotransporter 2 inhibitor, glucagon-like peptide type 1 receptor agonists, and nonsteroidal MRAs, it is unclear if the beneficial effect of AF ablation are additive on these pharmacotherapies. Finally, because the E/e' and systolic pulmonary pressures were not available as a metric of diastolic function, we replaced them with diastolic function grading, as reported in the trial for the calculation of a modified H₂FPEF score.

CONCLUSIONS

This post hoc analysis suggests a potential beneficial treatment effect of AF ablation in patients with a high probability for HFpEF (by a modified H₂FPEF score or

by echocardiography) on adverse clinical outcome, AF recurrence and functional status. Future studies are necessary investigating AF ablation in HFpEF.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Martens is supported by a grant from BAEF (Belgian American Educational Foundation) and by the Frans Van de Werf Fund. The BioLINCC (Biologic Specimen and Data Repository Information Coordinating Center) is funded by the National Institutes of Health. Dr Tang is a consultant for Sequana Medical, Cardiol Therapeutics Inc, Genomics plc, Zehna Therapeutics, Boston Scientific, WhiteSwell, Intellia Therapeutics, CardiaTec Biosciences, Bristol Myers Squibb, Alleviant Medical, Alexion Pharmaceuticals, Salubris Biotherapeutics, and BioCardia Inc; and has received honorarium from Springer, Belvoir Media Group, and the American Board of Internal Medicine. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr W.H. Wilson Tang, Kaufman Center for Heart Failure Treatment and Recovery, Department of Cardiovascular Medicine, Heart Vascular and Thoracic Institute, 9500 Euclid Avenue, Desk J3-4, Cleveland Clinic, Cleveland, Ohio 44195, USA. E-mail: tangw@ccf.org. X handle: [@WilsonTangMD](https://twitter.com/WilsonTangMD). OR Dr Pieter Martens, Department of Cardiology, Ziekenhuis Oost Limburg, Synapspark 1, 3600 Genk, Belgium. E-mail: Pieter_martens@icloud.com. X handle: [@PieterMartensMD](https://twitter.com/PieterMartensMD).

PERSPECTIVES

COMPETENCY IN PATIENT CARE AND PROCEDURAL

SKILLS: The presence of HFpEF at baseline (by H₂FPEF score or by echocardiography) was associated with significant benefits from AF ablation with a lower risk for adverse clinical outcomes.

TRANSLATIONAL OUTLOOK: These hypothesis-generating observations that warrant further validation studies that may help to prioritize those with elevated H₂FPEF score for early considerations for AF ablation.

REFERENCES

1. Joglar JA, Chung MK, Armbruster AL, et al. 2023 ACC/AHA/ACCP/HRs Guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2024;83: 109-279. <https://doi.org/10.1016/j.jacc.2023.08.017>
2. Black-Maier E, Ren X, Steinberg BA, et al. Catheter ablation of atrial fibrillation in patients with heart failure and preserved ejection fraction. *Heart Rhythm*. 2018;15:651-657. <https://doi.org/10.1016/j.hrthm.2017.12.001>
3. Di Biase L, Mohanty P, Mohanty S, et al. Ablation versus amiodarone for treatment of persistent atrial fibrillation in patients with congestive heart failure and an implanted device: results from the AATAC multicenter randomized trial. *Circulation*. 2016;133:1637-1644. <https://doi.org/10.1161/circulationaha.115.019406>
4. Fujimoto H, Doi N, Okayama S, et al. Long-term prognosis of patients undergoing radiofrequency catheter ablation for atrial fibrillation: comparison between heart failure subtypes based on left ventricular ejection fraction. *Europace*. 2022;24:576-586. <https://doi.org/10.1093/europace/euab201>
5. Hunter RJ, Berriman TJ, Diab I, et al. A randomized controlled trial of catheter ablation versus medical treatment of atrial fibrillation in heart failure (the CAMTAF trial). *Circ Arrhythm Electrophysiol*. 2014;7:31-38. <https://doi.org/10.1161/circep.113.000806>
6. Ichijo S, Miyazaki S, Kusa S, et al. Impact of catheter ablation of atrial fibrillation on long-term clinical outcomes in patients with heart failure. *J Cardiol*. 2018;72:240-246. <https://doi.org/10.1016/j.jjcc.2018.02.012>
7. Jones DG, Haldar SK, Hussain W, Sharma R, Francis DP, Rahman-Haley SL, McDonagh TA,

- Underwood SR, Markides V, Wong T. A randomized trial to assess catheter ablation versus rate control in the management of persistent atrial fibrillation in heart failure. *J Am Coll Cardiol*. 2013;61:1894-1903. <https://doi.org/10.1016/j.jacc.2013.01.069>
8. Marrouche NF, Brachmann J, Andresen D, et al. Catheter ablation for atrial fibrillation with heart failure. *N Engl J Med*. 2018;378:417-427. <https://doi.org/10.1056/NEJMoa1707855>
9. Sohns C, Fox H, Marrouche NF, et al. Catheter ablation in end-stage heart failure with atrial fibrillation. *N Engl J Med*. 2023;389:1380-1389. <https://doi.org/10.1056/NEJMoa2306037>
10. von Olshausen G, Benson L, Dahlström U, Lund LH, Savarese G, Braunschweig F. Catheter ablation for patients with atrial fibrillation and heart failure: insights from the Swedish Heart Failure Registry. *Eur J Heart Fail*. 2022;24:1636-1646. <https://doi.org/10.1002/ehf.2604>
11. Zylla MM, Leiner J, Rahm AK, et al. Catheter ablation of atrial fibrillation in patients with heart failure and preserved ejection fraction. *Circ Heart Fail*. 2022;15:e009281. <https://doi.org/10.1161/circheartfailure.121.009281>
12. Chiang D, Sugumar H, Segal L, et al. atrial fibrillation ablation for heart failure with preserved ejection fraction: a randomized controlled trial. *JACC Heart Fail*. 2023;11:646-658. <https://doi.org/10.1016/j.jchf.2023.01.008>
13. Melenovsky V, Hwang SJ, Redfield MM, Zakeri R, Lin G, Borlaug BA. Left atrial remodeling and function in advanced heart failure with preserved or reduced ejection fraction. *Circ Heart Fail*. 2015;8:295-303. <https://doi.org/10.1161/circheartfailure.114.001667>
14. Ariyaratnam JP, Elliott AD, Mishima RS, et al. Identification of subclinical heart failure with preserved ejection fraction in patients with symptomatic atrial fibrillation. *JACC Heart Fail*. 2023;11:1626-1638. <https://doi.org/10.1016/j.jchf.2023.07.019>
15. Reddy YNV, Carter RE, Obokata M, Redfield MM, Borlaug BA. A simple, evidence-based approach to help guide diagnosis of heart failure with preserved ejection fraction. *Circulation*. 2018;138:861-870. <https://doi.org/10.1161/circulationaha.118.034646>
16. Reddy YNV, Kaye DM, Handoko ML, et al. Diagnosis of heart failure with preserved ejection fraction among patients with unexplained dyspnea. *JAMA Cardiol*. 2022;7:891-899. <https://doi.org/10.1001/jamacardio.2022.1916>
17. Naser JA, Lee E, Scott CG, Kennedy AM, Pellikka PA, Lin G, Pislaru SV, Borlaug BA. Prevalence and incidence of diastolic dysfunction in atrial fibrillation: clinical implications. *Eur Heart J*. 2023;44:5049-5060. <https://doi.org/10.1093/eurheartj/ehad592>
18. Mark DB, Anstrom KJ, Sheng S, et al. Effect of catheter ablation vs medical therapy on quality of life among patients with atrial fibrillation: the CABANA randomized clinical trial. *JAMA*. 2019;321:1275-1285. <https://doi.org/10.1001/jama.2019.0692>
19. Packer DL, Mark DB, Robb RA, et al. Effect of catheter ablation vs antiarrhythmic drug therapy on mortality, stroke, bleeding, and cardiac arrest among patients with atrial fibrillation: the CABANA randomized clinical trial. *JAMA*. 2019;321:1261-1274. <https://doi.org/10.1001/jama.2019.0693>
20. Pieske B, Tschope C, de Boer RA, et al. How to diagnose heart failure with preserved ejection fraction: the HFA-PEFF diagnostic algorithm: a consensus recommendation from the Heart Failure Association (HFA) of the European Society of Cardiology (ESC). *Eur Heart J*. 2019;40:3297-3317. <https://doi.org/10.1093/eurheartj/ehz641>
21. Kirchhof P, Camm AJ, Goette A, et al. Early rhythm-control therapy in patients with atrial fibrillation. *N Engl J Med*. 2020;383:1305-1316. <https://doi.org/10.1056/NEJMoa2019422>
22. Kotecha D, Lam CS, Van Veldhuisen DJ, Van Gelder IC, Voors AA, Rienstra M. Heart failure with preserved ejection fraction and atrial fibrillation: vicious twins. *J Am Coll Cardiol*. 2016;68:2217-2228. <https://doi.org/10.1016/j.jacc.2016.08.048>
23. McDonagh TA, Metra M, Adamo M, et al. 2023 focused update of the 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2023;44:3627-3639. <https://doi.org/10.1093/eurheartj/ehad195>
24. Swedberg K, Zannad F, McMurray JJ, et al. Eplerenone and atrial fibrillation in mild systolic heart failure: results from the EMPHASIS-HF (Eplerenone in Mild Patients Hospitalization And Survival Study in Heart Failure) study. *J Am Coll Cardiol*. 2012;59:1598-1603. <https://doi.org/10.1016/j.jacc.2011.11.063>
25. Rienstra M, Hobbelt AH, Alings M, et al. Targeted therapy of underlying conditions improves sinus rhythm maintenance in patients with persistent atrial fibrillation: results of the RACE 3 trial. *Eur Heart J*. 2018;39:2987-2996. <https://doi.org/10.1093/eurheartj/ehx739>
26. Ariyaratnam JP, Mishima RS, Kadhim K, et al. Utility and validity of the HFA-PEFF and H(2)FPEF scores in patients with symptomatic atrial fibrillation. *JACC Heart Fail*. 2024;12:1015-1025. <https://doi.org/10.1016/j.jchf.2024.01.015>
27. Orati A, McIntyre WF, Parkash R, et al. Atrial fibrillation ablation in heart failure with reduced vs preserved ejection fraction: a systematic review and meta-analysis. *JAMA Cardiol*. 2024;9:545-555. <https://doi.org/10.1001/jamacardio.2024.0675>
28. Krishnamurthy A, Goyal P, Markowitz SM, et al. Outcomes of patients with heart failure with preserved ejection fraction undergoing catheter ablation of atrial fibrillation. *Heart Rhythm*. 2022;3:501-508. <https://doi.org/10.1016/j.hrroo.2022.06.012>
29. Packer DL, Piccini JP, Monahan KH, et al. Ablation versus drug therapy for atrial fibrillation in heart failure: results from the CABANA trial. *Circulation*. 2021;143:1377-1390. <https://doi.org/10.1161/circulationaha.120.050991>
30. Lindenfeld J, Costanzo MR, Zile MR, et al. Implantable hemodynamic monitors improve survival in patients with heart failure and reduced ejection fraction. *J Am Coll Cardiol*. 2024;83:682-694. <https://doi.org/10.1016/j.jacc.2023.11.030>

KEY WORDS atrial fibrillation, clinical trials, heart failure with preserved ejection fraction

APPENDIX For a supplemental figure, please see the online version of this paper.

